SUPPLY CHAIN TRACEABILITY ASSESSMENT

Yukigaya Chemical Industry Co., Ltd. (雪ヶ谷化学工業株式会社)

Feedstock verification and the critical upstream node

Japan

Subject	Yukigaya Chemical Industry Co., Ltd. (雪ヶ谷化学工業株式会社)
Jurisdiction	Japan
Case type	Supply-chain traceability assessment
Materials	NBR; SBR; natural rubber
Source basis	Public Japanese sources

Prepared from public sources only. No private, paid or non-public data was used.

1. Executive Research Question This investigation examines whether the publicly disclosed supply chain of natural-rubber cosmetic sponge products manufactured by Yukigaya Chemical Industry Co., Ltd. (雪ヶ谷化学工業株式会社) can be reconstructed beyond the Japanese manufacturer and its identified Thai upstream partner. The principal research question is: Can the publicly disclosed supply chain of Yukigaya Chemical Industry's natural-rubber cosmetic sponge products be reconstructed from the Japanese manufacturer back to the upstream raw-material source, and what can be independently verified about the companies and people involved? The investigation deliberately distinguishes between: 1. a company being publicly identified as a supplier; 2. a company actually processing or purchasing natural rubber; 3. a documented commercial relationship; 4. an identifiable upstream procurement network; 5. an identifiable individual farm, cooperative, trader, or producer; and 6. a demonstrably traceable physical origin of the rubber incorporated into Yukigaya's products. This distinction is essential because the existence of a documented supplier relationship does not, by itself, establish the geographic or individual agricultural origin of the raw material.

2. Scope and Methodology

The investigation covers the publicly accessible record concerning:

- Yukigaya Chemical Industry;
- its natural-rubber cosmetic products;
- Natural Art & Technology Co., Ltd. in Thailand;
- the historical entity Pan Asia Biotechnology Co., Ltd. ;
- relevant Thai rubber-processing and procurement structures;
- historical company records;
- industry directories;
- scientific publications identifying commercial rubber suppliers;
- Japanese corporate and sustainability disclosures;
- Thai corporate and industry information;
- farm-level and fair-trade disclosures;
- historical references that may have disappeared from current corporate websites. The research was conducted using a cross-source methodology rather than relying on a single

corporate narrative. Source hierarchy Evidence was assessed approximately in the following order: Tier 1 - Primary / quasi-primary evidence

- corporate disclosures;
- official government and registry records;
- historical company records;
- regulatory or industry documents;
- original scientific publications identifying commercial suppliers. Tier 2 - Independent institutional evidence
- industry associations;
- academic publications;
- specialist organizations;
- independent Japanese sustainability/ethical-business organizations. Tier 3 - Commercial databases and directories
- business directories;
- industry catalogues;
- historical commercial listings. Tier 4 - Inferential evidence
- geographic proximity;
- shared addresses;
- overlapping personnel;

- product compatibility;
- historical temporal continuity;
- common industrial ecosystem. Inferential evidence was not treated as equivalent to direct documentary evidence.

3. Material Definition

The investigation concerns natural rubber (NR) . Natural rubber was kept analytically separate from synthetic and alternative elastomers, including:

- NBR;
- SBR;
- polyurethane;
- other synthetic rubber compounds. This distinction is important because Yukigaya manufactures a broader range of rubber-related products, and the presence of the word "rubber" in a product description is not sufficient to establish that natural rubber is involved.

The relevant supply-chain question therefore concerns the natural-rubber component/material associated with Yukigaya's cosmetic sponge products , rather than Yukigaya's entire product portfolio.

4. Yukigaya Chemical Industry -

Corporate and Product Baseline

Yukigaya Chemical Industry is a Japanese manufacturer operating in the rubber-products sector. Its publicly disclosed activities include products incorporating natural rubber, including cosmetic sponge products. The company has publicly discussed the sourcing of natural rubber in the context of sustainability, fair-trade principles, producer conditions, and supply-chain traceability. This is particularly relevant because Yukigaya does not describe its natural-rubber sourcing solely as an ordinary industrial procurement activity. Its public materials describe efforts to investigate conditions at the agricultural production level. The company has publicly discussed field investigations involving Thai rubber-producing areas and has described checks concerning issues including:

- child labour;
- forced labour;
- working conditions;
- environmental considerations;
- remuneration received by producers. These disclosures establish that Yukigaya's own traceability model extends conceptually beyond the immediate industrial supplier. However, this does

not automatically mean that the individual farms or farmers are publicly identifiable . That distinction becomes central later in the investigation.

5. Yukigaya's Farm-Level Verification

Claims

One of the strongest pieces of evidence relevant to the research question is Yukigaya's repeated disclosure that personnel in Thailand have conducted investigations at rubber-producing farms. The company has described personnel visiting farms and conducting interviews and field checks concerning the conditions under which natural rubber is produced. Later sustainability disclosures indicate that this was not necessarily a one-off exercise. Yukigaya continued reporting farm-level investigations in subsequent years, including assessments involving labour and environmental conditions. This produces an important evidence distinction: What is established Yukigaya has publicly represented that it performs farm-level investigation / verification activities in Thailand. What is not established by those disclosures The public material reviewed does not provide a complete list of:

- individual farms;
- individual farmers;
- farm coordinates;
- specific collection stations;
- specific cooperatives;
- individual upstream traders;
- purchase volumes attributable to each farm;
- transaction records connecting a named farm to a specific batch of material used by Yukigaya. Therefore, the existence of farm-level verification should not be confused with public farm-level traceability.

6. The Critical Upstream Node: Natural

Art & Technology

The investigation identifies Natural Art & Technology Co., Ltd. (NAT) as the principal Thai upstream entity relevant to Yukigaya's natural-rubber supply chain. The importance of NAT is not based merely on geographical proximity. Multiple strands of evidence place NAT within the natural-rubber processing and procurement chain. NAT is associated with the Rayong rubber-processing ecosystem and has been independently identified in scientific and commercial sources as a supplier/producer of natural-rubber materials. Scientific publications have identified Natural Art & Technology as a commercial source of natural-rubber latex used in research. For example, academic literature has documented researchers purchasing natural-rubber latex from: Natural Art & Technology Co., Ltd., Rayong, Thailand. Such records are significant because they independently establish that NAT is not merely a company appearing in a Yukigaya document: it is an operational natural-rubber supplier whose products were commercially available to external users.

7. Historical Continuity: Pan Asia Biotechnology → Natural Art & Technology

Historical research significantly strengthens the identification of NAT as an established rubber-industry node. Earlier records identify: Pan Asia Biotechnology Co., Ltd. as a Rayong-based natural-rubber company. Historical industry material places Pan Asia Biotechnology at: 91 Moo 1, Laharn, Pluak Daeng, Rayong, Thailand and identifies it as dealing in natural-rubber products including:

- concentrated latex;
- STR grades;
- other natural-rubber materials. Historical industry material also identifies a company contact associated with Pan Asia Biotechnology, providing an additional historical corporate identifier. This is important because it permits the investigation to move beyond a simple current-company

lookup. The evidence indicates continuity between the historical Pan Asia Biotechnology operation and the later Natural Art & Technology entity. The historical record also demonstrates that the operation was already commercially involved with natural-rubber materials well before the current Yukigaya sourcing disclosures.

8. Independent Evidence of Pan Asia's Rubber Activities

The historical Pan Asia identification is supported by academic literature. Scientific publications from the period before the transition to the Natural Art & Technology name identify Pan Asia Biotechnology Co., Ltd., Rayong, Thailand as a source of natural-rubber materials. Among the materials identified in scientific publications are:

- concentrated natural-rubber latex;
- high-ammonia natural-rubber latex;
- skim natural-rubber latex;
- STR-grade natural rubber. This provides an important historical continuity: PAN ASIA BIOTECHNOLOGY

│ │ documented natural-rubber production/supply │ ▼ Natural-rubber latex / STR products │ │ corporate continuity ▼ NATURAL ART & TECHNOLOGY │ │ documented natural-rubber production/supply ▼ Natural-rubber latex / related products The evidence therefore supports treating Pan Asia Biotechnology not as an unrelated company with a similar profile, but as a historically relevant predecessor identity in the reconstruction of the Thai rubber-processing node.

9. What Can Be Established About NAT's

Position in the Supply Chain?

At this stage, the evidence supports the following proposition: Natural Art & Technology operated as a natural-rubber processing/procurement entity capable of supplying natural-rubber latex and related materials to downstream users. This conclusion is supported independently of Yukigaya's own claims. The available evidence includes: 1. historical industry records identifying Pan Asia Biotechnology as a natural-rubber company; 2. scientific publications identifying Pan Asia as a commercial source of natural-rubber materials; 3. subsequent records identifying Natural Art & Technology as a natural-rubber supplier; 4. scientific publications independently purchasing natural-rubber latex from NAT; 5. commercial information describing NAT's rubber-related activities. The evidence therefore establishes NAT as a genuine operational node in the natural-rubber industry. It does not yet establish the identity of NAT's own raw-material suppliers . That distinction is fundamental.

10. The First Major Evidence Boundary

The investigation can currently reconstruct the supply chain to approximately this level: YUKIGAYA CHEMICAL INDUSTRY │ │ documented natural-rubber sourcing relationship ▼ NATURAL ART & TECHNOLOGY │ │ documented natural-rubber processing/procurement ▼ UPSTREAM LATEX PROCUREMENT NETWORK │ ├── FARMERS ? ├── COOPERATIVES ? ├── COLLECTION CENTRES ? ├── TRADERS ? └── LOCAL PROCESSORS ? The question marks are deliberate. At this point in the investigation, the existence of an upstream procurement layer is strongly supported, but the publicly available evidence does not yet permit individual upstream nodes to be assigned confidently to NAT.

11. Why This Distinction Matters

There are several possible supply-chain structures between a Thai rubber processor and an individual farmer. For example: Farm ↓ Collection centre ↓ Trader ↓ Processor ↓ NAT or: Farm ↓ Cooperative ↓ NAT or: Multiple farms ↓ Local collection network ↓ Trader / processor ↓ NAT Yukigaya's public statements concerning farm visits demonstrate that the company is concerned with conditions at the producer level. They do not, by themselves, establish which of these procurement structures applies to the specific latex purchased by NAT. Consequently, the investigation does not infer a farm-to-NAT relationship solely from the fact that rubber farms exist in the surrounding region.

12. Preliminary Corporate Finding

At the end of this stage, the evidence supports the following findings: Yukigaya Chemical Industry Status: Verified Japanese manufacturer and relevant downstream entity. Natural-rubber involvement: Confirmed. Relevant sustainability/traceability activity: Confirmed through company disclosures. Farm-level investigation claims: Confirmed as a publicly disclosed activity. Natural Art & Technology Status: Verified Thai natural-rubber industry entity. Operational relevance: Confirmed. Natural-rubber/latex activity: Independently supported. Historical continuity with Pan Asia Biotechnology: Strongly supported. Position as an upstream industrial node for Yukigaya's relevant supply chain: Supported by the case evidence. Individual upstream producer Identity: Not established. Specific cooperative Identity and relationship to NAT: Not established. Specific collection centre Identity and relationship to NAT: Not established. Specific trader Identity and relationship to NAT: Not established. Physical agricultural origin of the rubber incorporated into the investigated Yukigaya products Not yet independently established.

13. Interim Assessment

The investigation has therefore established a credible and independently corroborated intermediate supply-chain node , rather than a complete farm-to-manufacturer chain. The strongest defensible reconstruction at this stage is: YUKIGAYA CHEMICAL INDUSTRY │ │ ▼ NATURAL ART & TECHNOLOGY │ │ ▼ NATURAL-RUBBER / LATEX PROCUREMENT NETWORK │ ▼ UNIDENTIFIED UPSTREAM SOURCES The key unresolved question is no longer whether NAT is a genuine natural-rubber company. That question is substantially answered. The unresolved question is: Who supplied NAT with the natural rubber or latex that subsequently entered the material stream relevant to Yukigaya? Answering that question requires moving below the processor level and testing specific Thai suppliers, cooperatives, collection centres, traders, historical procurement records, geographic clusters, and individual producer networks. That constitutes the next stage of the investigation.

Supply Chain Traceability Assessment:

Yukigaya Chemical Industry

Part II - Historical Continuity, Corporate Network &

Upstream Investigation

14. Historical Reconstruction of the Thai

Upstream Node

The current Natural Art & Technology entity cannot be assessed adequately without reconstructing its historical corporate identity. The historical record identifies Pan Asia Biotechnology Co., Ltd. as an earlier entity operating in the same natural-rubber industry and geographic area. This historical layer is important because older records contain information that is no longer readily visible in current corporate material.

14.1 Pan Asia Biotechnology

Historical industry records identify Pan Asia Biotechnology as a Rayong-based natural-rubber company. A 2011 industry directory records: Pan Asia Biotechnology Co., Ltd. Address: 91 Moo 1, Laharn, Pluak Daeng, Rayong 21140, Thailand The same historical entry identifies the company as dealing in natural-rubber products including:

- concentrated latex;
- STR XL;
- STR 5CV;
- other natural-rubber grades. The record also identifies a historical company contact: Chanchira Tongsom

and telephone information associated with the company. This provides several independent identifiers for historical research: Pan Asia Biotechnology │ ├── 91 Moo 1 │ Laharn │ Pluak Daeng │ Rayong │ ├── Natural-rubber products │ ├── Historical telephone number │ └── Chanchira Tongsom These identifiers are significantly more useful for historical reconstruction than the company name alone.

15. Scientific Evidence of Pan Asia's Historical Operations

The historical corporate identification is supported by scientific publications. Academic researchers independently identified Pan Asia Biotechnology as a commercial source of natural-rubber materials. The documented materials include:

- concentrated natural-rubber latex;
- high-ammonia natural-rubber latex;
- skim natural-rubber latex;
- STR-grade natural rubber. This is significant because these publications were not produced to document Yukigaya's supply chain. They represent independent downstream users identifying their commercial source of rubber

material. The evidence therefore establishes that Pan Asia was functioning as a genuine natural-rubber supplier rather than merely appearing in a corporate directory.

16. Corporate Continuity: Pan Asia → Natural Art & Technology

The historical evidence supports the following corporate timeline: 2000s │ │ Pan Asia Biotechnology │ │ Natural-rubber processing / supply │ ▼ 2011 │ │ Pan Asia documented at │ 91 Moo 1, Laharn, │ Pluak Daeng, Rayong │ ▼ 2012 │ │ Pan Asia independently documented │ as supplier of natural-rubber latex │ ▼ 2013 │ │ Corporate transition │ ▼ Natural Art & Technology │ │ Natural-rubber processing / supply │ ▼ Present The exact legal mechanics of every historical corporate change should be distinguished from the broader operational continuity. The evidence is strongest for business/activity continuity . It is weaker for reconstructing every intermediate corporate event, shareholder change, or management transition. Therefore: Pan Asia Biotechnology → Natural Art & Technology is treated as a strongly supported historical continuity, while individual undocumented corporate events are not inferred.

17. Why the Historical Layer Matters for Traceability

The historical material substantially expands the available search surface. A current search limited to: "Natural Art & Technology supplier" can miss relevant records. Historical searches using:

- Pan Asia Biotechnology;
- old addresses;
- old telephone numbers;
- historical personnel;
- historical product descriptions; produce independent evidence of the same rubber-processing ecosystem. This is particularly important for upstream research because older procurement relationships

may have disappeared from current websites while remaining visible in:

- scientific publications;
- industry directories;
- historical catalogues;
- archived company pages;
- export documentation;

- industry association material. The investigation therefore treats historical corporate identity as an investigative pivot rather than merely background information.

18. Current Operational Position of

Natural Art & Technology

Natural Art & Technology is documented as operating within the Rayong natural-rubber industry. Independent scientific literature identifies NAT as a commercial supplier of natural-rubber latex. One study, for example, identifies natural-rubber latex purchased from: Natural Art & Technology Co., Ltd., Rayong, Thailand. Other scientific work similarly identifies NAT as a source of natural-rubber latex. This provides independent evidence that NAT's output is not merely theoretical or based solely on Yukigaya's description. It also establishes a useful material-level connection: NAT │ ├── Natural-rubber latex │ ├── Concentrated / ammonia latex │ └── Other natural-rubber materials However, the material identity still needs to be kept separate from the specific material used in a particular Yukigaya cosmetic sponge. The evidence demonstrates that NAT handles relevant natural-rubber materials. It does not independently disclose the batch-level or farm-level origin of the rubber incorporated into a particular Yukigaya product.

19. Geographic Concentration: Pluak

Daeng, Rayong

The historical and current evidence repeatedly places the relevant industrial activity in: Pluak Daeng District, Rayong Province, Eastern Thailand. This is not sufficient to establish the agricultural origin of the rubber. It does, however, establish a meaningful geographic context for the investigation. The area contains an established rubber-processing and agricultural ecosystem, including rubber plantations, collection structures, cooperatives and processors. The distinction is: Processing geography ≠ Agricultural origin Therefore, the location of NAT's facility cannot be used as evidence that the rubber itself originated in Pluak Daeng. It can only be used to identify a relevant procurement ecosystem for further investigation.

20. Candidate Upstream Node: Pluak

Daeng Rubber Cooperative

A particularly relevant candidate discovered during the upstream investigation is a rubber cooperative operating in Pluak Daeng: Pluak Daeng Rubber Plantation Fund Cooperative The cooperative is publicly associated with rubber production and handling, including rubber sheet and liquid-rubber activities. Its location is within the same district as the historical Pan Asia/NAT operation. This creates a legitimate investigative lead: Pluak Daeng rubber-producing network │ │ ▼ Pluak Daeng Rubber Cooperative │ │ ? ▼ Natural Art & Technology However, no reviewed source establishes a direct commercial relationship between this cooperative and NAT. Accordingly, the cooperative is classified as: Potential upstream ecosystem node - not a verified NAT supplier. Confidence Low for the specific NAT relationship. High for the cooperative's existence and participation in the local rubber ecosystem.

21. Historical Rubber-Industry Directory

Evidence

A historical Thai natural-rubber industry directory provides an additional geographic clue. The directory identifies rubber-processing organizations in Rayong, including operations in the Pluak Daeng area. One listed rubber-processing/cooperative node is associated with: Ta Sit / Pluak Daeng, Rayong. The documented activities include processing of:

- fresh rubber;
- rubber sheets;
- cup lump. This demonstrates that the immediate geographical environment surrounding NAT contained multiple potential upstream rubber collection and processing nodes. Again, this is

ecosystem evidence , not transaction evidence. The analytical distinction is critical: A local supplier candidate becomes an upstream hypothesis only when a source establishes a relationship with NAT. Without such evidence, geographic proximity cannot substitute for procurement documentation.

22. Rayong Bangkok Rubber - A

Significant Historical Lead

Another important result emerged from historical industry records. The same historical industry material identifies: Rayong Bangkok Rubber Co., Ltd. at an address recorded as: 91 Moo 1, Laharn, Pluak Daeng, Rayong The address substantially overlaps with the historical address recorded for Pan Asia Biotechnology. This creates a potentially significant corporate-network lead: 91 Moo 1 Laharn Pluak Daeng Rayong ├── Pan Asia Biotechnology │ └── Rayong Bangkok Rubber The two entities are also independently associated with natural-rubber latex. This makes the overlap more interesting than a generic address coincidence. Nevertheless, the available evidence does not establish:

- common ownership;
- common management;
- subsidi status;
- a joint venture;
- a supplier relationship;
- a customer relationship;
- a corporate merger;
- or a direct relationship between Rayong Bangkok Rubber and NAT. Therefore, the shared-address observation must remain a

lead rather than a conclusion .

23. Rayong Bangkok Rubber as an Independent Latex Node

Scientific publications independently identify Rayong Bangkok Rubber as a source of natural-rubber latex. Researchers have reported obtaining commercial high-ammonia natural-rubber latex from the company. This independently establishes: Rayong Bangkok Rubber │ ▼ Natural-rubber latex The fact that both Pan Asia and Rayong Bangkok Rubber appear independently in scientific literature as commercial rubber suppliers confirms that the historical Rayong industrial ecosystem contained multiple commercially active latex suppliers. It does not , however, establish that Rayong Bangkok Rubber supplied Pan Asia or NAT.

24. Historical Personnel: Chanchira Tongsom

The historical Pan Asia record identifies: Chanchira Tongsom as a company contact. This person is therefore a useful historical corporate identifier. However, the investigation deliberately avoids treating name similarity as person identification. No sufficient evidence was found establishing that this individual:

- owned NAT;
- owned Pan Asia;
- managed NAT;
- controlled a rubber cooperative;
- represented a specific upstream farm;
- or maintained a documented procurement relationship with a particular producer. Accordingly, Chanchira Tongsom should be treated as:

Verified historical Pan Asia contact; upstream relationship unverified. This is an example of the investigation's identity-control rule: A name match is not sufficient to establish a person/entity relationship.

25. Yukigaya's Local Verification Network

Yukigaya's own disclosures provide another important component of the network. The company describes local personnel participating in investigations of rubber-producing farms. Independent Japanese reporting has described the mechanism in terms of:

- Thailand-based personnel;
- cooperation with local entities;
- visits to production farms;
- interviews and field investigation. This establishes that Yukigaya's traceability model appears to reach beyond the immediate industrial supplier. However, the publicly disclosed information does not identify all intermediaries between

Yukigaya and the farms visited. A plausible structure therefore remains: Yukigaya │ │ Thai/local personnel │ ▼ Local coordination / supplier network │ ▼ Rubber-producing farms The identity of the intermediate entities remains unresolved.

26. A Source Discrepancy Requiring

Caution

Some secondary descriptions characterize the people conducting farm investigations as employees associated with a Thai factory. Other materials, including material closer to Yukigaya's own wording, describe Thailand-based or stationed personnel and cooperation with local companies/farms. This distinction matters. It would be inappropriate to convert the wording: "Thai factory staff" into an assumption that the staff belonged specifically to NAT. The more conservative interpretation is: Yukigaya had personnel in Thailand who participated in farm-level investigations, potentially in cooperation with local companies or production entities. Whether those local entities were NAT itself, another supplier, an intermediary, or an agricultural organization has not been independently established. This ambiguity is retained rather than silently resolved.

27. Upstream Procurement Model

The available evidence supports the existence of a procurement layer between agricultural production and NAT. The most defensible generic model is: INDIVIDUAL FARMERS │ ▼ LOCAL COLLECTION / COOPERATIVE NETWORK │ ▼ TRADER / INTERMEDIARY │ ▼ NATURAL ART & TECHNOLOGY │ ▼ PROCESSED NATURAL RUBBER / LATEX │ ▼ YUKIGAYA However, this is a structural model , not a verified transaction chain. The individual components below NAT remain unidentified in the public evidence reviewed.

28. Cross-Link Analysis

The investigation specifically tested whether indirect evidence could bridge this gap. The principal cross-link categories were: Geographic overlap

- NAT / historical Pan Asia - Pluak Daeng, Rayong.
- Rubber cooperatives - Pluak Daeng.
- Other rubber-processing companies - Pluak Daeng/Rayong. Result: Supports a shared industrial ecosystem, but not a specific commercial relationship.

Product compatibility

- NAT - natural-rubber latex.
- Candidate cooperatives/processors - natural-rubber products.
- Yukigaya - natural-rubber products. Result: Technically compatible, but insufficient to prove procurement. Historical continuity
- Pan Asia - natural-rubber processing/supply.
- Natural Art & Technology - same industry and geographic ecosystem. Result:

Strong support for corporate/operational continuity. Personnel overlap Historical company contacts were investigated. Result: No sufficiently strong person-to-person upstream link established. Address overlap Pan Asia and Rayong Bangkok Rubber appear at the same historical address. Result: Significant lead, but no verified corporate relationship. Farm-level Yukigaya activity Yukigaya independently reports farm visits and producer-level investigations. Result: Strong evidence that farm-level verification occurs, but farm identities are not publicly disclosed.

29. Evidence Matrix

Relationship / Finding Evidence Assessment Yukigaya uses natural rubber Corporate/product disclosures Confirmed Yukigaya conducts Thai farm-level investigations Yukigaya / Japanese institutional material Confirmed NAT is a natural-rubber/latex company Scientific + commercial sources Confirmed Pan Asia was a natural-rubber supplier Historical industry + scientific sources Confirmed Pan Asia and NAT represent historical business continuity Corporate/history evidence Probable-High NAT operates in the Pluak Daeng/Rayong rubber ecosystem Corporate/industry records Confirmed Pluak Daeng contains rubber cooperatives Thai industry records Confirmed Specific cooperative supplies NAT No direct evidence found Unverified Rayong Bangkok Rubber supplied NAT No direct evidence found Unverified Rayong Bangkok Rubber and Pan Asia shared an address Historical directory Confirmed as recorded Shared address means common ownership No sufficient evidence Unverified Specific farm supplies NAT No direct evidence found Unverified Specific farmer supplies Yukigaya No direct evidence found Unverified

30. Interim Upstream Reconstruction

At the end of this stage, the strongest evidence-supported structure is: YUKIGAYA │ │ documented sourcing relationship ▼ NATURAL ART & TECHNOLOGY │ │ documented natural-rubber │ processing / procurement ▼ UPSTREAM PROCUREMENT LAYER │ ┌──────────┬──────────┐ │ │ │ ▼ ▼ ▼ FARMERS COOPERATIVES TRADERS │ │ └──────────┴──────────┘ ▼ NATURAL RUBBER The following nodes have been identified as investigative leads , but not established links: Pluak Daeng Rubber Cooperative │ └──── ? ──── NAT Rayong Bangkok Rubber │ └──── ? ──── NAT / Pan Asia These hypotheses remain separate from the verified chain.

31. What the Investigation Has

Established So Far

The evidence is now sufficient to make several relatively strong statements. First, Yukigaya's natural-rubber sourcing is not limited in its publicly described verification model to the Japanese manufacturer or its immediate supplier. The company reports activity at the agricultural level. Second, Natural Art & Technology is a genuine natural-rubber industry node independently documented by sources outside Yukigaya. Third, the historical Pan Asia Biotechnology identity provides a substantial additional layer of evidence demonstrating long-term rubber-processing activity in the same Rayong/Pluak Daeng industrial environment. Fourth, the local Rayong ecosystem contains multiple rubber-processing, collection and cooperative structures capable in principle of supplying natural-rubber processors. However, none of these findings establishes the final missing link: specific upstream producer → NAT → Yukigaya The agricultural source therefore remains outside the publicly verifiable chain.

32. Principal Unresolved Question

The remaining investigative problem is no longer: "Does NAT actually handle natural rubber?" The available evidence strongly supports that proposition. It is: Which specific farms, cooperatives, collection centres, traders, or other raw-material suppliers provided the natural rubber/latex acquired by NAT during the period relevant to Yukigaya's products? This is the critical evidentiary gap that determines whether the supply chain can be reconstructed all the way to the physical raw-material source. The next stage must therefore focus primarily on contradiction testing, negative evidence, supplier identification attempts, historical procurement traces, and determining whether the public evidence contains any hidden cross-link capable of closing this final gap.

Supply Chain Traceability Assessment:

Yukigaya Chemical Industry

Part III - Contradiction Testing, Final Reconstruction

& Assessment

33. Independent Verification and

Contradiction Testing

The final stage of the investigation was designed not to extend the supply-chain model, but to test whether the reconstructed chain could be falsified, weakened, or materially altered by contradictory evidence. Particular attention was given to:

- conflicting company names;
- historical corporate identities;
- inconsistent addresses;
- inconsistent descriptions of the relevant material;
- conflicting descriptions of Yukigaya's Thai personnel;
- possible alternative suppliers;
- corporate and personnel cross-links;
- geographic discrepancies;
- independent evidence concerning candidate upstream entities. The purpose was to distinguish between a genuinely traceable chain and a chain that appears complete only because unsupported assumptions have been inserted between documented nodes.

34. Material Identity Test

The first contradiction test concerns the material itself. The investigated material is natural rubber / natural-rubber latex . The available evidence independently identifies:

- Yukigaya's relevant products as involving natural rubber;
- NAT as a natural-rubber processor/supplier;
- historical Pan Asia operations as involving natural-rubber latex and STR products;
- independent downstream users purchasing natural-rubber latex from NAT/Pan Asia. No evidence reviewed requires replacing the relevant material with:
- NBR;
- SBR;
- polyurethane;
- or another synthetic elastomer. Therefore, the central material classification is internally consistent. Finding

Material identity: CONFIRMED.

35. Corporate Identity Test

The investigation separately tested whether Natural Art & Technology might simply be a company with a similar name rather than the actual Thai rubber-processing entity relevant to the case. The evidence does not support that interpretation. NAT is independently associated with:

- Rayong;
- natural-rubber latex;
- commercial rubber supply;
- scientific users purchasing its latex. Historical records independently identify Pan Asia Biotechnology in the same general operational context and at the relevant historical address. Finding

NAT identity as a genuine natural-rubber industry entity: CONFIRMED.

36. Pan Asia / NAT Historical Continuity

The historical record provides substantial support for continuity between the Pan Asia Biotechnology operation and the later Natural Art & Technology identity. The continuity is supported by the combination of:

- historical company name;
- geographic location;
- natural-rubber product profile;
- historical industry records;
- scientific publications;
- subsequent NAT activity. However, the investigation does not claim that every corporate event between the two names has been independently reconstructed. In particular, the available public evidence is insufficient to establish every:
- shareholder transition;
- director transition;
- corporate restructuring;
- legal transaction;
- ownership transfer. Accordingly, the appropriate classification is: Strongly supported operational/corporate continuity; individual undocumented corporate

events remain unverified. Finding Pan Asia → NAT continuity: HIGH / PROBABLE-HIGH.

37. Yukigaya → NAT Relationship

The evidence supports NAT as the relevant Thai upstream industrial node in the investigated Yukigaya supply chain. The relationship is supported by Yukigaya's own sourcing disclosures and by independent evidence establishing NAT's actual natural-rubber operations. The independent evidence is important because it prevents the relationship from resting solely on Yukigaya's corporate narrative. NAT is independently documented as a commercial source of natural-rubber latex. Finding Yukigaya → NAT: CONFIRMED / HIGH. The exact commercial mechanics of individual transactions - including:

- purchase orders;
- shipment volumes;
- batch numbers;
- individual invoices; are not publicly available in the evidence reviewed. Therefore, the relationship is confirmed at the company/supply-chain level, not reconstructed

transaction-by-transaction.

38. NAT → Upstream Supplier Test

This is the decisive test. The investigation attempted to identify the entities immediately upstream of NAT using:

- historical Pan Asia records;
- Thai rubber industry directories;
- cooperative records;
- local rubber-processing entities;
- commercial directories;
- scientific publications;
- historical addresses;
- telephone identifiers;
- historical personnel;
- geographic cross-links;
- candidate rubber cooperatives;
- candidate latex suppliers. Despite these searches, no public document was identified that states, in substance: Company/Farm/Cooperative X supplied natural rubber or latex to NAT/ Pan Asia.

This is the critical negative finding.

39. Candidate: Pluak Daeng Rubber

Cooperative

The existence of a rubber cooperative in the Pluak Daeng area is independently established. Its activities are compatible with the type of agricultural rubber supply required by a processor such as NAT. The geographic relationship is also meaningful because:

- NAT is associated with Pluak Daeng;
- historical Pan Asia records place the operation there;
- rubber cultivation and processing are established in the surrounding region. However, no source reviewed establishes:

Pluak Daeng Rubber Cooperative → NAT as a commercial relationship. Therefore, the hypothesis cannot be promoted into the verified supply chain. Classification Entity existence: Confirmed. Rubber-sector role: Confirmed. Relationship to NAT: Unverified. Confidence that it was an actual NAT supplier: Low.

40. Candidate: Rayong Bangkok Rubber

Rayong Bangkok Rubber represents a more unusual historical lead. Historical industry records place Rayong Bangkok Rubber at an address substantially overlapping with the historical Pan Asia address: 91 Moo 1, Laharn, Pluak Daeng, Rayong. Rayong Bangkok Rubber is independently documented as a natural-rubber latex supplier. This creates a potentially relevant historical corporate-network hypothesis. However, no sufficiently strong evidence was found establishing:

- common ownership;
- common directors;
- common corporate control;
- merger;
- subsidiary relationship;
- supply relationship;
- purchase relationship;
- or direct operational integration with Pan Asia/NAT. Consequently: Classification

Rayong Bangkok Rubber existence: Confirmed. Natural-rubber activity: Confirmed. Historical address overlap: Confirmed as recorded. Relationship with Pan Asia/NAT: Unverified. Possible relevance: Investigative lead only. This entity must therefore not be presented in the final supply chain as a verified supplier.

41. Candidate: Individual Historical

Contact

The historical Pan Asia record identifies a specific company contact. The investigation attempted to use this individual as a cross-link to other companies or upstream actors. No sufficiently strong evidence was found establishing that the person:

- owned an upstream supplier;
- controlled a cooperative;
- represented a farm;
- held a verified management position at NAT during the relevant period;
- or personally participated in the procurement chain documented here. This demonstrates why the investigation does not use simple name matching as evidence of

identity or control. Finding Historical corporate contact: Confirmed. Upstream role: Unverified.

42. Farm-Level Traceability Test

Yukigaya's public disclosures are unusually important at this stage. The company reports farm-level investigation and producer-level verification in Thailand. This establishes that the company has, at least according to its own disclosures, visibility extending beyond the immediate industrial supplier. However, the publicly available information reviewed does not disclose a complete farm-level dataset. Specifically, the investigation did not identify publicly verifiable records connecting the investigated Yukigaya products to:

- a named farm;
- a named farmer;
- a farm coordinate;
- a named collection station;
- a named cooperative;
- a specific transaction;

- a specific batch. Therefore: Farm-level inspection ≠ publicly reconstructable farm-level supply chain. This distinction is central to the final conclusion.

43. Geographic Origin Test

The investigation established a strong connection between the relevant industrial node and Rayong / Pluak Daeng . However, this cannot be converted into a claim that the natural rubber used by Yukigaya originated in Pluak Daeng. Natural-rubber processors can procure raw material from wider regional networks. Possible supply areas include other parts of Eastern Thailand and potentially other rubber-producing regions. No reviewed evidence provides a sufficiently specific geographic origin for the particular natural rubber entering the investigated Yukigaya products. Therefore: Established Processing/procurement geography: Rayong / Pluak Daeng. Not established Agricultural origin of the material used by Yukigaya. This distinction prevents a common OSINT error: inferring raw-material origin from the location of the processor.

44. Negative Evidence Summary

The absence of the following records is itself an important finding after repeated targeted searches. No sufficiently reliable public evidence was located for:

- NAT's named upstream supplier list;
- NAT's supplier contracts;
- purchase orders identifying farms;
- invoices connecting named farms to NAT;
- public procurement records identifying NAT's latex suppliers;
- named collection stations supplying NAT;
- named cooperatives documented as NAT suppliers;
- individual farmers documented as NAT suppliers;
- farm coordinates associated with Yukigaya's relevant product supply;
- batch-level traceability records;
- publicly accessible supplier codes linking NAT to specific farms;
- an independent document connecting a named farm directly to Yukigaya's natural-rubber material. This does

not prove that such records do not exist privately. It establishes only that they were not found in the publicly accessible evidence reviewed.

45. Alternative Explanations

The evidence gap should not automatically be interpreted as concealment or misconduct. Several structurally plausible explanations remain. Hypothesis A - Multi-tier procurement NAT may procure through: Farm ↓ Collection point ↓ Trader ↓ NAT In such a model, the immediate supplier may itself aggregate material from numerous farms. Hypothesis B - Cooperative procurement The structure could instead be: Multiple farmers ↓ Cooperative ↓ NAT Hypothesis C - Mixed procurement NAT may use multiple channels simultaneously: Farm ──────┐ Cooperative ├──→ NAT Trader ──────┘ Hypothesis D - Yukigaya has additional visibility Yukigaya may possess farm-level information that is simply not publicly disclosed. The company's repeated farm-investigation disclosures are consistent with this possibility. None of these explanations can currently be selected as the definitive procurement structure.

46. Assessment of the Major Hypotheses

Hypothesis Evidence Assessment NAT purchases natural-rubber/latex material Independent commercial/scientific evidence HIGH Pan Asia was historical predecessor/continuous operation Historical + scientific + geographic evidence HIGH / PROBABLE-HIGH Yukigaya's natural-rubber chain reaches Thailand Corporate disclosures + supplier evidence HIGH Yukigaya performs farm-level verification Repeated company disclosures HIGH Upstream supply includes farmers/collection structures Industry structure + material requirements HIGH as structural proposition Pluak Daeng cooperative supplies NAT Geographic/product compatibility only LOW / UNVERIFIED Rayong Bangkok Rubber supplied NAT Address overlap + same industry LOW / UNVERIFIED A specific named farmer supplies NAT No direct evidence UNVERIFIED Rubber used by Yukigaya originated in Pluak Daeng No direct geographic evidence UNVERIFIED Specific farm can be linked to Yukigaya's product No public transaction/batch evidence UNVERIFIED

47. Final Reconstructed Supply Chain

The maximum chain that can be reconstructed without introducing unsupported assumptions is: ┌──────────────────────────────────────┐ │ YUKIGAYA CHEMICAL INDUSTRY │ │ Japan │ └──────────────────────────────────────┘ │ │ HIGH │ documented sourcing relationship ▼ ┌──────────────────────────────────────┐ │ NATURAL ART & TECHNOLOGY │ │ Rayong, Thailand │ └──────────────────────────────────────┘ │ │ HIGH │ natural-rubber processing / │ latex procurement ▼ ┌──────────────────────────────────────┐ │ UPSTREAM PROCUREMENT NETWORK │ │ │ │ Exact participants unknown │ └──────────────────────────────────────┘ │ ┌──────────────┐ │ │ ▼ ▼ ▼ Farms Co-ops Traders │ │ └──────────────┘ ▼ NATURAL RUBBER SOURCE The chain becomes evidentially discontinuous below NAT. The correct representation is therefore not : Yukigaya ↓ NAT ↓ Cooperative X ↓ Farm Y ↓ Farmer Z because Cooperative X, Farm Y and Farmer Z cannot be independently established as participants in this specific chain.

48. Maximum Defensible Traceability

Level

The investigation therefore establishes a partial but meaningful traceability chain . Level 1 - Manufacturer Yukigaya Chemical Industry Status: HIGH The relevant Japanese manufacturer and natural-rubber product context are established. Level 2 - Thai industrial supplier/processor Natural Art & Technology Status: HIGH The company is independently documented as a natural-rubber/latex industry entity. Level 3 - Raw-material procurement layer NAT upstream procurement network Status: MEDIUM/HIGH as an operational layer The existence of such a procurement layer is strongly supported by the nature of NAT's operations and its documented handling of natural-rubber latex. Level 4 - Specific upstream supplier Status: UNVERIFIED No specific supplier can be established with sufficient evidence. Level 5 - Cooperative / collection point Status: UNVERIFIED Candidate entities exist, but no direct relationship with NAT was established. Level 6 - Farm Status: UNVERIFIED Yukigaya reports farm-level investigations, but the public evidence does not identify the relevant farms sufficiently to connect them to the investigated product supply. Level 7 - Individual producer Status: UNVERIFIED No individual producer could be independently linked to the investigated supply chain.

49. Confidence Framework

The final evidence classification is: Dimension Finding Confidence Yukigaya identity Japanese manufacturer HIGH Natural-rubber product relevance Confirmed HIGH Yukigaya → Thai supply chain Documented HIGH Dimension Finding Confidence NAT identity Thai natural-rubber entity HIGH NAT natural-rubber activity Independent evidence HIGH Pan Asia historical identity Confirmed HIGH Pan Asia → NAT continuity Strongly supported HIGH / PROBABLE-HIGH NAT → upstream procurement layer Operationally established MEDIUM/HIGH Specific NAT supplier Not identified UNVERIFIED Pluak Daeng cooperative → NAT No direct evidence LOW / UNVERIFIED Rayong Bangkok Rubber → NAT No direct evidence LOW / UNVERIFIED Specific geographic origin of Yukigaya rubber Not established UNVERIFIED Specific farm → NAT Not established UNVERIFIED Specific farmer → Yukigaya Not established UNVERIFIED

50. Answer to the Main Research

Question

The original research question was: Can the publicly disclosed supply chain of Yukigaya Chemical Industry's natural-rubber cosmetic sponge products be reconstructed from the Japanese manufacturer back to the upstream raw-material source, and what can be independently verified about the companies and people involved?

Answer

Only partially. The publicly available evidence is sufficient to reconstruct a credible and independently corroborated chain from: Yukigaya Chemical Industry → Natural Art & Technology and to establish that NAT operates as a natural-rubber/latex processing and procurement node in Rayong, Thailand. The historical record further supports the reconstruction of the Thai industrial node through the earlier Pan Asia Biotechnology identity. Yukigaya's own disclosures additionally establish that its traceability programme extends to farm-level investigations in Thailand. However, the public evidence reviewed does not permit the chain to be reconstructed with sufficient confidence from NAT to a specifically identified:

- upstream supplier;
- cooperative;
- collection centre;
- trader;
- farm;
- or individual producer. Consequently, the physical agricultural origin of the natural rubber entering the investigated Yukigaya product supply chain cannot currently be independently established from the public record reviewed .

51. What Can Be Independently Verified

About the Companies?

Yukigaya Chemical Industry The following are independently supportable:

- Japanese corporate identity;
- natural-rubber product involvement;
- Thai natural-rubber sourcing activity;
- farm-level investigation claims;
- sustainability/traceability programme. Natural Art & Technology The following are independently supportable:
- existence as a Thai rubber-industry entity;
- Rayong geographic association;
- natural-rubber/latex activities;
- commercial supply of natural-rubber latex;
- historical connection to the Pan Asia Biotechnology business identity. Pan Asia Biotechnology

The following are independently supportable:

- historical existence;
- Rayong/Pluak Daeng location;
- natural-rubber/latex activities;
- commercial supply to downstream users;
- historical company personnel/contact information. Candidate upstream entities The existence of some rubber-sector entities in the relevant geographic ecosystem can be independently verified. Their

commercial relationship to NAT cannot .

52. What Can Be Independently Verified

About the People?

The investigation successfully identified historical corporate personnel/contact information associated with the Pan Asia entity. However, the evidence does not support extending those identities into the upstream agricultural network. In particular, no individual can be responsibly classified as: "the farmer supplying NAT" or: "the producer whose rubber entered Yukigaya's cosmetic sponge supply" on the basis of the public evidence reviewed. This is intentional. The investigation does not convert:

- name similarity;
- geographic proximity;
- company association;
- or industry participation into individual identification.

53. Final Negative Finding

The most important negative finding of the investigation is therefore: The publicly disclosed supply chain terminates, for independent verification purposes, at the Thai industrial procurement/processing level represented by Natural Art & Technology. The evidence indicates that deeper traceability may exist operationally. Yukigaya's repeated farm-level investigations strongly suggest that the company has some access to producer-level information. But the public record does not expose enough identifiers to independently reproduce that portion of the chain. This is the difference between: internal traceability and publicly reproducible traceability . The first may exist. The second could not be demonstrated beyond NAT.

54. Final Assessment

Confirmed

- Yukigaya is the relevant Japanese manufacturer.
- Natural rubber is materially relevant to the investigated product category.
- Yukigaya has publicly documented Thai natural-rubber sourcing and farm-level verification activities.
- Natural Art & Technology is a genuine Thai natural-rubber/latex industry entity.
- NAT's natural-rubber activities are independently corroborated.
- Pan Asia Biotechnology provides a strong historical layer for reconstructing the Thai rubber-processing node.
- Rayong/Pluak Daeng is a documented processing/procurement environment for natural rubber. Probable / strongly supported
- Pan Asia Biotechnology and Natural Art & Technology represent historical continuity of the relevant Thai rubber-processing business.
- NAT obtains natural-rubber material through an upstream procurement network involving agricultural production and/or intermediaries. Possible but unverified
- Specific Pluak Daeng cooperatives may have participated in NAT's procurement ecosystem.
- Rayong Bangkok Rubber may have had a historical corporate or operational relationship with Pan Asia/NAT because of the address overlap and common industry context. Unverified
- Any specific NAT upstream supplier.
- Any specific cooperative supplying NAT.
- Any specific collection station supplying NAT.
- Any specific farm supplying NAT.
- Any individual farmer supplying the material.
- The precise province/district/farm of origin of the natural rubber incorporated into the investigated Yukigaya products.

55. Final Conclusion

The investigation does not support a claim that the complete natural-rubber supply chain of Yukigaya's cosmetic sponge products can be reconstructed publicly down to the individual agricultural source. It does support a substantially narrower and defensible conclusion: The publicly available evidence allows the supply chain to be reconstructed with high confidence from Yukigaya Chemical Industry to Natural Art & Technology in Rayong, Thailand, with strong historical evidence connecting the latter's operation to Pan Asia Biotechnology. Independent sources confirm NAT's role in natural-rubber/latex processing and supply, while Yukigaya independently documents farm-level verification activities. However, the public evidence reviewed does not identify and independently connect the specific upstream farms, cooperatives, collection centres, traders, or individual producers whose natural rubber entered the NAT material stream relevant to Yukigaya. Accordingly, the final traceability assessment is: PARTIALLY RECONSTRUCTABLE - HIGH CONFIDENCE TO THE NAT PROCESSING/PROCUREMENT NODE; UNVERIFIABLE FROM PUBLIC SOURCES AT THE SPECIFIC AGRICULTURAL SOURCE LEVEL. The appropriate endpoint of the public evidence is therefore: YUKIGAYA CHEMICAL INDUSTRY │ │ HIGH ▼ NATURAL ART & TECHNOLOGY │ │ HIGH ▼ NATURAL-RUBBER PROCUREMENT NETWORK │ │ ▼ [redacted] PUBLIC EVIDENCE GAP [redacted] │ ▼ Specific farm / producer UNVERIFIED This boundary should be treated as a finding of the investigation , not as an invitation to fill the missing section with inference. Any future attempt to close the remaining gap would require evidence not presently available in the public record reviewed, such as procurement documentation, supplier disclosures, certification records containing supplier-level identifiers, transaction records, farm identifiers, or direct documentary confirmation from the relevant parties.